

A complete proof of the shooting characterisation: the noisy-Airy inner exit measure is Tracy–Widom

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Abstract

We prove that the inner exit measure of the noisy folded limit cycle—the law of the first-node level Y_{node} of the swept noisy Airy equation—coincides with the law of minus the ground-state eigenvalue of the stochastic Airy operator, and hence with Tracy–Widom: $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta) \stackrel{d}{=} \text{TW}_\beta$, $\eta = 2/\sqrt{\beta}$. The proof is elementary given the random-Schrödinger framework of Ramírez–Rider–Virág (RRV): the swept solution is the *energy-zero* solution of a random Schrödinger operator M , its first zero is the level at which the truncated M has ground state 0, and that level is read off through a *monotone spectral function* whose defining event involves only a *deterministic* translation—so the stationarity and reflection symmetry of white noise close the identity. The single nontrivial random shift, which obstructed earlier arguments, is bypassed by Dirichlet domain monotonicity.

1 Setup and standing facts

The two objects. Fix $\eta > 0$ and $\beta = 4/\eta^2$. Let $\tilde{\xi}$ be a white noise on $\mathbb{R}$ (the distributional derivative of a two-sided Brownian motion W).

Object 1 (the inner exit level). In the rescaling chart the noisy canard solves, in the slow variable Y ,

$$u_{YY} = (Y - \eta \tilde{\xi}) u. \quad (1)$$

(This is (1) because $Y = Y_0 - T$ gives $\partial_Y^2 = \partial_T^2$, and the time white noise becomes a white noise in Y .) Let ψ be the solution of (1) that is recessive (square-integrable) as $Y \rightarrow +\infty$; its largest zero is the *inner exit level* Y_{node} .

Object 2 (the SAO ground state). The stochastic Airy operator

$$H_\beta = -\frac{d^2}{dx^2} + x + \frac{2}{\sqrt{\beta}} b'(x) \quad \text{on } [0, \infty), \text{ Dirichlet at } 0, \quad (2)$$

(b' a white noise, $2/\sqrt{\beta} = \eta$) has a.s. a simple lowest eigenvalue $\Lambda_0(\beta)$.

The operator M . Write (1) as $Mu = 0$ with the random Schrödinger operator

$$M = -\frac{d^2}{dY^2} + Y - \eta \tilde{\xi}(Y), \quad (3)$$

i.e. ψ is the energy-0 recessive solution of M . For $\ell \in \mathbb{R}$ let M_ℓ denote M acting on $L^2([\ell, \infty))$ with a Dirichlet condition at ℓ .

We use four standing facts, each classical in its area.

Standing fact 1 (white-noise invariances). For every deterministic $c \in \mathbb{R}$, $\tilde{\xi}(c + \cdot) \stackrel{d}{=} \tilde{\xi}$ (stationarity) and $-\tilde{\xi} \stackrel{d}{=} \tilde{\xi}$ (reflection symmetry). Consequently $-\eta \tilde{\xi}(c + \cdot) \stackrel{d}{=} \frac{2}{\sqrt{\beta}} b'$.

Standing fact 2 (random Schrödinger operator). For a.e. realisation and every ℓ , M_ℓ is self-adjoint and bounded below with purely discrete spectrum near its bottom; its lowest eigenvalue is simple with a ground-state eigenfunction that is strictly positive on (ℓ, ∞) . The same holds for H_β . (*This is the RRV construction [1]; the confining potential $Y \rightarrow +\infty$ and the white-noise term are handled by their quadratic-form/Riccati analysis. See also [2] for 1D Schrödinger operators with white-noise potential.*)

Standing fact 3 (Weyl limit point at $+\infty$). Since the potential $\rightarrow +\infty$, equation (1) is in the limit-point case at $+\infty$: for each energy there is a unique-up-to-scale solution that is square-integrable near $+\infty$ [3]. In particular the recessive solution ψ exists and is unique up to scale.

Standing fact 4 (RRV edge). $-\Lambda_0(\beta) \stackrel{d}{=} \text{TW}_\beta$ [1].

2 The theorem

Theorem 1 (Shooting characterisation). *With the notation above,*

$$Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta) \stackrel{d}{=} \text{TW}_\beta, \quad \eta = 2/\sqrt{\beta}.$$

Moreover, if u^{Y_0} denotes the solution of (1) started at $Y = Y_0$ on the canard branch ($u_Y/u = -\sqrt{Y_0}$) and $Y_{\text{node}}(Y_0)$ its largest zero, then $Y_{\text{node}}(Y_0) \rightarrow Y_{\text{node}}$ almost surely as $Y_0 \rightarrow \infty$; hence $Y_{\text{node}}(Y_0) \xrightarrow{d} -\Lambda_0(\beta)$.

The proof is four lemmas. The crux is Lemma 3: the event $\{Y_{\text{node}} \leq t\}$ equals $\{G(t) \geq 0\}$ for a monotone spectral function G , and the level t there is *deterministic*, so Fact 1 applies.

Lemma 1 (the exit level is well defined). *Almost surely ψ exists (Fact 3), is real and not identically zero, is positive for all large Y , and has a largest zero $Y_{\text{node}} \in \mathbb{R}$.*

Proof. Existence and uniqueness up to scale are Fact 3; take ψ real (the equation has real coefficients). For Y large the potential $Y - \eta \tilde{\xi}$ is positive, so by (1) ψ is convex where positive and concave where negative; the recessive (decaying) branch is positive for all large Y . For $Y \rightarrow -\infty$ the potential $\rightarrow -\infty$ and ψ oscillates with infinitely many zeros, which are isolated (a nontrivial solution of a second-order linear ODE has only simple, isolated zeros). The set of zeros is therefore bounded above (none for large Y) and discrete, so it has a maximum Y_{node} . $\square$

Lemma 2 (forgetting: the sweep limit). *Almost surely $Y_{\text{node}}(Y_0) \rightarrow Y_{\text{node}}$ as $Y_0 \rightarrow \infty$, exponentially fast.*

Proof. Work with the Riccati variable $P = u_Y/u$, which solves $P_Y = (Y - \eta \tilde{\xi}) - P^2$. On the barrier $\{Y > 0\}$ the two WKB branches are $P \approx \pm \sqrt{Y}$, and linearising about $P = -\sqrt{Y}$ gives $\partial_P[(Y - \eta \tilde{\xi}) - P^2] = -2P \approx +2\sqrt{Y}$. Thus along *decreasing* Y the branch $P = -\sqrt{Y}$ is attracting: if $P^{(1)}, P^{(2)}$ are two solutions with $\delta = P^{(1)} - P^{(2)}$ then $\delta_Y = -(P^{(1)} + P^{(2)})\delta$, so for Y decreasing from Y_0 ,

$$|\delta(Y)| = |\delta(Y_0)| \exp\left(-\int_Y^{Y_0} (P^{(1)} + P^{(2)}) dY'\right) \leq |\delta(Y_0)| e^{-2 \int_Y^{Y_0} \sqrt{Y'} (1+o(1)) dY'}.$$

The canard datum $P^{Y_0}(Y_0) = -\sqrt{Y_0}$ differs from the recessive value $P_\psi(Y_0)$ by the WKB error $\delta(Y_0) = O(Y_0^{-1})$ (Fact 3); integrating down to any fixed Y_* contracts this to $O(Y_0^{-1}e^{-\frac{4}{3}Y_0^{3/2}})$, uniformly. Hence $P^{Y_0} \rightarrow P_\psi$ locally uniformly on $\{Y \leq Y_*\}$, and the largest zero $Y_{\text{node}}(Y_0)$ (a continuous functional of P on a neighbourhood of Y_{node} , where $P \rightarrow -\infty$ transversally) converges to Y_{node} . $\square$

Lemma 3 (monotone spectral function). *Let $G(\ell)$ be the lowest eigenvalue of M_ℓ (Fact 2). Almost surely:*

- (i) G is strictly increasing on $\mathbb{R}$;
- (ii) $G(Y_{\text{node}}) = 0$;
- (iii) for every $t \in \mathbb{R}$, $\{Y_{\text{node}} \leq t\} = \{G(t) \geq 0\}$.

Proof. (i) For $\ell_1 < \ell_2$, the form domain of M_{ℓ_2} embeds into that of M_{ℓ_1} by extension by zero, and the quadratic forms agree on the image; hence by the min-max principle $G(\ell_1) \leq G(\ell_2)$ (Dirichlet domain monotonicity [4, XIII]). The inequality is strict: the ground state φ_1 of M_{ℓ_1} is strictly positive on (ℓ_1, ∞) (Fact 2), so $\varphi_1(\ell_2) > 0$ and φ_1 is *not* admissible for M_{ℓ_2} (which requires vanishing at ℓ_2); the variational minimum therefore strictly increases.

(ii) Restrict ψ to $[Y_{\text{node}}, \infty)$. There $\psi(Y_{\text{node}}) = 0$ (Dirichlet satisfied), $\psi \in L^2$ (recessive decay at $+\infty$), $M\psi = 0$, and ψ has no zero in $(Y_{\text{node}}, \infty)$ since Y_{node} is the *largest* zero (Lemma 1). A nodeless L^2 eigenfunction is the ground state (Fact 2), so 0 is the lowest eigenvalue of $M_{Y_{\text{node}}}$: $G(Y_{\text{node}}) = 0$.

(iii) By (i) and (ii), $G(t) \geq 0 = G(Y_{\text{node}}) \iff t \geq Y_{\text{node}}$. $\square$

Lemma 4 (deterministic translation). *For each fixed $t \in \mathbb{R}$, $G(t) \stackrel{d}{=} \Lambda_0(\beta) + t$.*

Proof. Fix t and substitute the *deterministic* shift $Y = t + x$, $x \geq 0$. Then

$$M_t = -\frac{d^2}{dx^2} + (t + x) - \eta \tilde{\xi}(t + x) = \underbrace{\left[-\frac{d^2}{dx^2} + x - \eta \tilde{\xi}(t + x)\right]}_{=: H'_t} + t,$$

an operator on $[0, \infty)$ with Dirichlet at 0. By Fact 1, $\tilde{\xi}(t + \cdot) \stackrel{d}{=} \tilde{\xi}$ and $-\eta \tilde{\xi}(t + \cdot) \stackrel{d}{=} \frac{2}{\sqrt{\beta}} b'$, so $H'_t \stackrel{d}{=} H_\beta$ as random operators; in particular their lowest eigenvalues have the same law, $\Lambda_0(H'_t) \stackrel{d}{=} \Lambda_0(\beta)$. Since $M_t = H'_t + t$, $G(t) = \Lambda_0(H'_t) + t$, whence $G(t) \stackrel{d}{=} \Lambda_0(\beta) + t$. $\square$

Proof of Theorem 1. Fix $t \in \mathbb{R}$. Using Lemma 3(iii) (a pathwise event identity) and then Lemma 4 (a statement about the law of $G(t)$ at the fixed level t),

$$\mathbb{P}(Y_{\text{node}} \leq t) = \mathbb{P}(G(t) \geq 0) = \mathbb{P}(\Lambda_0(\beta) + t \geq 0) = \mathbb{P}(\Lambda_0(\beta) \geq -t) = \mathbb{P}(-\Lambda_0(\beta) \leq t).$$

As t was arbitrary, $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta)$; by Fact 4, $Y_{\text{node}} \stackrel{d}{=} \text{TW}_\beta$. The sweep statement is Lemma 2 together with the continuous-mapping theorem. $\square$

3 Discussion

What is elementary and what is cited. Given Facts 1–4, the proof is elementary: Lemmas 1, 3 and 4 use only Sturm theory, Dirichlet domain monotonicity, and the substitution $Y = t + x$; Lemma 2 is a Gronwall estimate on the Riccati. The only heavy input is Fact 2—the rigorous meaning of a Schrödinger operator with a white-noise potential and the simplicity/positivity of its ground state. That is precisely the RRV framework we are matching to, applied to M , which is an SAO *in law* (Lemma 4 at $t = 0$); so the proof imports no machinery beyond what the statement already presupposes.

Why earlier arguments stalled. A direct “reflection” argument tries to translate by the random level Y_{node} , producing a noise $\tilde{\xi}(Y_{\text{node}} + \cdot)$ shifted by a noise-dependent amount—not a clean white noise. Lemma 3 removes this obstruction: the event $\{Y_{\text{node}} \leq t\}$ is recast as $\{G(t) \geq 0\}$ with t *deterministic*, after which Fact 1 applies verbatim.

Numerical confirmation. (i) The conclusion $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta)$ is confirmed by computing $-\Lambda_0$ as the smallest eigenvalue of the tridiagonal discretisation of H_β and Y_{node} by ODE shooting—two methods with no shared code path—which agree in mean, variance and skewness at $\beta = 1, 2, 4$, and match the exact TW_β densities. (ii) The mechanism of Lemma 3 is confirmed directly: for fixed realisations, $G(\ell)$ (smallest eigenvalue of the discretised M_ℓ) is monotone increasing and crosses 0 at the recessive solution’s largest zero Y_{node} computed from the same noise.

Consequence. The inner exit measure of the noisy folded limit cycle is TW_β with $\beta = 4/\eta^2$, exactly; the early-escape probability is the tail mass $1 - F_\beta(0)$, and the folded-limit-cycle canard escape thereby lies in the Tracy–Widom / KPZ edge universality class.

References

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